



S5301 HAL Final Inspection Form
(Ref. "Final Inspection Procedure" PRO-102026-EN-xx)

1. Test Information

1. Simulator Serial No.: _____	4. Type: New ☐ Repair ☐ Upgrade ☐ Show ☐ Loaner ☐
2. UNI version: _____	5. Color: Light ☐ Medium ☐ Dark ☐
3. HAL S5301 version: _____	6. Power Adapter AC Voltage: _____

2. Cosmetic

Visual Inspection	Pass	Fail	5. Removable Inserts (continued):	Pass	Fail
1. Skin Condition	☐	☐	• Humerus IO (upper left shoulder)	☐	☐
2. ECG & Defibrillation Snaps	☐	☐	• Intramuscular (IM) (upper left arm)	☐	☐
3. Glucose Finger (left index finger)	☐	☐	• Antecubital Fossa (inside left elbow)	☐	☐
4. Simulator S/N Label (under abdomen insert)	☐	☐	• IBP/Drug Recognition (left forearm)	☐	☐
5. Removable Inserts:			• Left IV Hand (top left hand)	☐	☐
• Tracheostomy (neck)	☐	☐	• Right IV (right forearm)	☐	☐
• Pneumothorax (upper left chest)	☐	☐	• Right IV Hand (top right hand)	☐	☐
• Chest Tube (left chest side)	☐	☐	• Tibia IO (lower right leg)	☐	☐
• Abdominal (healthy abdomen)	☐	☐	• Chest overlay	☐	☐
• Abdominal wound (lower abdomen)	☐	☐	• Male Genitalia	☐	☐

Notes: _____

3. Connection with UNI

Connection Types	Pass	Fail
1. RF Radio	☐	☐
2. Wired	☐	☐
3. Bluetooth	☐	☐

Notes: _____

Note: Use Bluetooth connection with UNI for the remaining tests.

4. Hardware

Fill and Check Reservoirs for Leaks	Pass	Fail	Battery	Pass	Fail	Status (%)
1. Fluid Reservoir (urinary port)	☐	☐	1. Battery Status w/o charger	☐	☐	_____
2. Blood Reservoir (abdomen port)	☐	☐	2. Battery with charger	☐	☐	_____

Notes: _____

5. Urine

Urine Functions	Pass	Fail	Urine (continued)	Pass	Fail
1. Voluntary Urine – On/Off	☐	☐	4. Catheterization:		
2. Urine Rate (250 ml/hr)	☐	☐	• UNI logs catheterization	☐	☐
3. Blood Rate (250 ml/hr)	☐	☐	• Activated only when detect catheter	☐	☐

Notes: _____



6. Cardiac / Osat

Cardiac Functions	Pass	Fail	Cardiac Functions (continued)	Pass	Fail
1. Artificial Ventilation with BVM Mask	☐	☐	12. Sternum/Apex Pacing Detection	☐	☐
2. Auto Lung Sound	☐	☐	13. Sternum/Apex Pacing Capture	☐	☐
3. Chest Compression	☐	☐	14. Cardioversion Detection	☐	☐
4. Auto Carotid Pulse	☐	☐	15. Cardioversion Capture	☐	☐
5. ECG Rhythm at ECG Sites	☐	☐	16. Defibrillation Shock Detection	☐	☐
6. ECG Rhythm at Shock Sites	☐	☐	17. Defibrillation Shock Capture	☐	☐
7. Rhythms and Heart Rates OK	☐	☐			
8. Respiratory Graph	☐	☐	Oxygen Saturation	Pass	Fail
9. 12 Lead ECG (print out)	☐	☐	1. O2 Sat. Calibration	☐	☐
10. AP Pacing Detection	☐	☐	2. O2 Sat. Reading (left mid. finger – 98/80/60 %)	☐	☐
11. AP Pacing Capture	☐	☐			

Notes: _____

7. Airway

Airway	Pass	Fail
1. Tongue Edema – On/Off	☐	☐
2. Laryngospasm – On/Off	☐	☐
3. Pharyngeal Swelling – On/Off	☐	☐

Notes: _____

8. Circulation

Manual Blood Pressure	Pulses	Pass	Fail
BP = 120/80	1. Carotid Right Pulse (right neck)	☐	☐
Using Stethoscope/Blood Pressure Cuff and Korotkoff (Sound)	2. Carotid Left Pulse (left neck)	☐	☐
Systolic (120) Diastolic (80)	3. Brachial Right Pulse (inner right bicep)	☐	☐
Left Arm _____	4. Brachial Left Pulse (inner left bicep)	☐	☐
	5. Femoral Right Pulse (right hip)	☐	☐
	6. Femoral Left Pulse (left hip)	☐	☐
	7. Popliteal Right Pulse (back of right knee)	☐	☐
	8. Popliteal Left Pulse (back of left knee)	☐	☐
	9. Radial Pulse (wrist):		
	• Left – On/Off	☐	☐
	• Right – On/Off	☐	☐
	10. Pedal Pulse (foot):		
	• Left – On/Off	☐	☐
	• Right – On/Off	☐	☐

Notes: _____



9. Lower Arms / Vascular

IBP (Invasive Blood Pressure)				Drug Recognition (Left Forearm)		Pass	Fail
Left Arm (if Trial 1 fails, complete Trial 2 and 3):				1. Program drug A		☐	☐
BP Value	Trial 1	Trial 2	Trial 3	2. Program drug B		☐	☐
200/120 (+/-10)	_____	_____	_____	3. Volume detection		☐	☐
160/90 (+/-10)	_____	_____	_____	Detects 10 ml (+/- 2) (if Trial 1 fails, complete Trials 2 and 3):			
120/80 (+/-10)	_____	_____	_____	Drug	Trial 1	Trial 2	Trial 3
100/60 (+/-10)	_____	_____	_____	A	_____	_____	_____
				B	_____	_____	_____
Train of Four				Vascular Access		Pass	Fail
Use of Neuromuscular Transmission Monitor (NTM) on left thumb.				1. Right IV Forearm – Drain Port		☐	☐
(if Trial 1 fails, complete Trials 2 and 3):				2. Right tibia I/O with drainage (air test)		☐	☐
TOF Value	Trial 1	Trial 2	Trial 3	3. Left humeral I/O with I/O changes (air test)		☐	☐
100 (110 - 98)	_____	_____	_____	4. Blood draw		☐	☐
90 (95 - 88)	_____	_____	_____				
80 (85 - 78)	_____	_____	_____				
70 (75 - 68)	_____	_____	_____				

Notes: _____

10. Breathing

Chest Rise		Pass	Fail	Intubation		Pass	Fail
1. Left – On/Off		☐	☐	1. ET Tube Position 3 (Too Deep)		☐	☐
2. Right – On/Off		☐	☐	2. Right Mainstem (right lung only)		☐	☐
3. Pneumothorax – On/Off		☐	☐	3. ET Tube Position 1 (Normal)		☐	☐
4. Respiratory Rate (13/30 /min)		☐	☐	4. Bilateral chest rise ventilation		☐	☐
Fluid Chest Tube (Hemothorax)		Pass	Fail	5. ET Tube Position 0 (ET Tube Removed)		☐	☐
1. Hemothorax – On/Off		☐	☐	6. Nasal Intubation		☐	☐
2. Blood flow (250 ml/min)		☐	☐	7. Esophageal Intubation Detection		☐	☐
3. Fluid flow (150 ml/min)		☐	☐	8. Ventilation with iGEL		☐	☐

Notes: _____

11. Neurological / CRT

Arm Movement - Right		Pass	Fail	Finger Pressure		Pass	Fail
1. Clavicle 100%		☐	☐	1. Right Middle Finger		☐	☐
2. Clavicle 0% (Decerebrate)		☐	☐				
3. Handshake (grip the simulator's hand)		☐	☐	Capillary Refill Time		Pass	Fail
4. Squeeze Hand (grip the simulator's hand)		☐	☐	1. CRT – 5 sec. (right middle finger)		☐	☐
5. Relaxed		☐	☐				
6. Trapezius Pinch (left clavicle)		☐	☐				
7. Sternum Rub (apply pressure to sternum)		☐	☐				

Notes: _____



12. Trauma / CO2

Warning - Turn simulator OFF when removing or attaching limbs to the body.

Right Arm Bleed

Pass Fail

1. Right Arm Bleed – On/Off ☐ ☐
2. Tourniquet stops bleeding ☐ ☐

Left Leg Bleed

Pass Fail

1. Left Leg Bleed – On/Off ☐ ☐
2. Tourniquet stops bleeding ☐ ☐

Abdomen Bleed

Pass Fail

1. Abdomen Bleed – On/Off ☐ ☐
2. Abdominal packing stops bleeding ☐ ☐

Lung CO2

Pass Fail

1. Lung CO2 (0-10) ☐ ☐

Notes: _____

13. Seizures / Supraorbital

Seizures

Pass Fail

1. Non Tonic Clonic ☐ ☐
2. Generalized Tonic Clonic ☐ ☐
3. None ☐ ☐

Supraorbital Notch

Pass Fail

1. Left and Right – Both ☐ ☐
2. Left and Right – Disabled ☐ ☐

Notes: _____

14. Cephalic / Speech

Skin Appearance

Pass Fail

1. Cyanosis (blue) ☐ ☐
2. Pale (white) ☐ ☐
3. Redness (red) ☐ ☐
4. Normal (no color) ☐ ☐

Idle Movement

Pass Fail

1. Anxious ☐ ☐
2. None ☐ ☐

Mouth Movement

Pass Fail

1. Closed ☐ ☐
2. Open ☐ ☐
3. Teeth Chattering ☐ ☐
4. Relaxed ☐ ☐

Neck Movement (Yaw)

Pass Fail

1. Center ☐ ☐
2. Sideways – Left / Right ☐ ☐
3. Track Visual ☐ ☐
4. Track Audio ☐ ☐
5. Shaking (1 sec) ☐ ☐
6. Relaxed ☐ ☐

Neck Movement (Pitch)

Pass Fail

1. Center ☐ ☐
2. Shaking (1 sec) ☐ ☐
3. Vertical – Front/Back ☐ ☐
4. Relaxed ☐ ☐

Tears

Pass Fail

1. Left (9) ☐ ☐
2. Right (9) ☐ ☐

Sweat

Pass Fail

1. Left (9) ☐ ☐
2. Right (9) ☐ ☐

Speech

Pass Fail

1. Streaming Audio with Headphone ☐ ☐
2. Streaming Audio with Microphone ☐ ☐
3. Duplex Streaming Audio ☐ ☐
4. Pre-recorded speech ☐ ☐
5. Mouth moves with speech ☐ ☐
6. Voice recognition to have a conversation ☐ ☐
7. Response to questions and commands ☐ ☐

Notes: _____



15. Facial Expressions (Manual Control)

Facial Expressions	Pass	Fail	Nystagmus Options	Pass	Fail
1. Transient Pain	☐	☐	1. Downbeat (vertical joystick control)	☐	☐
2. Ongoing Pain	☐	☐	2. Upbeat (vertical joystick control)	☐	☐
3. Scared	☐	☐	3. Right beat (horizontal joystick control)	☐	☐
4. Lethargic	☐	☐	4. Left beat (horizontal joystick control)	☐	☐
5. Quizzical	☐	☐	5. Horizontal	☐	☐
6. Worried	☐	☐	6. Vertical	☐	☐
7. Smile	☐	☐	7. Pendular	☐	☐
8. Sad	☐	☐	8. Right Wink	☐	☐
9. Right Facial Droop	☐	☐	9. Left Wink	☐	☐
10. Left Facial Droop	☐	☐			
11. None	☐	☐			

Face Designer	Pass	Fail
1. Eyebrows – Right and Left	☐	☐
2. Eyelids – Right and Left	☐	☐
3. Lips – Right and Left	☐	☐
4. Jaw	☐	☐

Notes: _____

16. Ocular

Blink State	Pass	Fail	Eye Auto Tracking	Pass	Fail
1. Open	☐	☐	1. Right eye only	☐	☐
2. Closed	☐	☐	2. Left eye only	☐	☐
3. 20 blinks/min	☐	☐	3. Both eyes	☐	☐
			4. Horizontal nystagmus	☐	☐
			5. Vertical nystagmus	☐	☐
			6. Right nystagmus	☐	☐
			7. Left nystagmus	☐	☐
			8. Disabled	☐	☐

Eyelid State	Pass	Fail	Eye Strabismus	Pass	Fail
1. Left – Open, Close, Normal	☐	☐	1. Right Esotropia	☐	☐
2. Right – Open, Close, Normal	☐	☐	2. Right Exotropia	☐	☐
			3. Left Esotropia	☐	☐
			4. Left Exotropia	☐	☐
			5. Crossed eyes	☐	☐
			6. None	☐	☐

Pupil Dilation	Pass	Fail	Eye Movement	Pass	Fail
1. Left (2) / Right (9)	☐	☐	1. Left	☐	☐
2. Left (9) / Right (2)	☐	☐	2. Right	☐	☐
			3. Up	☐	☐
			4. Down	☐	☐
			5. None	☐	☐

Pupil Reaction	Pass	Fail
1. Left – On/Off	☐	☐
2. Right – On/Off	☐	☐

Pupil Reaction Time	Pass	Fail
1. Reaction Time (5 sec)	☐	☐

Consensual Reaction To Light	Pass	Fail
1. Consensual Reaction To Light – On/Off	☐	☐

Notes: _____



16. Ocular (continued)

Blepharospasm	Pass	Fail	Ptosis Severity	Pass	Fail
1. Left – On/Off	☐	☐	1. Left (0 and 5)	☐	☐
2. Right – On/Off	☐	☐	2. Right (0 and 5)	☐	☐

Notes: _____

17. Sounds (Enable Auscultation for this section.)

Lung Sounds - Anterior	Pass	Fail	Bowel Sounds	Pass	Fail
1. Upper Left Lung Sounds	☐	☐	1. Upper Left Bowel	☐	☐
2. Upper Right Lung Sounds	☐	☐	2. Upper Right Bowel	☐	☐
3. Lower Left Lung Sounds	☐	☐	3. Lower Left Bowel	☐	☐
4. Lower Right Lung Sounds	☐	☐	4. Lower Right Bowel	☐	☐
Lung Sounds - Posterior	Pass	Fail	Heart Sounds	Pass	Fail
1. Upper Left Lung Sounds	☐	☐	1. Heart Sound Aortic	☐	☐
2. Upper Right Lung Sounds	☐	☐	2. Heart Sound Pulmonic	☐	☐
3. Lower Left Lung Sounds	☐	☐	3. Heart Sound Tricuspid	☐	☐
4. Lower Right Lung Sounds	☐	☐	4. Heart Sound Mitral	☐	☐
Throat Sounds	Pass	Fail			
1. Throat Sounds	☐	☐			

Notes: _____

18. Ultrasound

Ultrasound Probe	Pass	Fail	Ultrasound Probe (continued)	Pass	Fail
1. Probe Gyro	☐	☐	3. Detects simulator's ultrasound sensors:		
2. Probe Pressure on Simulator	☐	☐	• Neck (1 sensor)	☐	☐
			• Belly Button (1 sensor)	☐	☐
			• Left/Right Nipples (2 sensors)	☐	☐
			• Left/Right sides (4 sensors on each side)	☐	☐

Notes: _____

19. Lung Compliance / Resistance

Variable Lung Unit Compliance				Lung Resistance			
PEEP: 0 mmHg; f: 13; V: 500 ml				PEEP: 0 mmHg; f: 13; V: 500 ml			
LC Level	Trial 1	Trial 2	Trial 3	LR Level	Trial 1	Trial 2	Trial 3
5	_____	_____	_____	0	_____	_____	_____
4	_____	_____	_____	1	_____	_____	_____
3	_____	_____	_____	3	_____	_____	_____
2	_____	_____	_____	5	_____	_____	_____
1	_____	_____	_____	7	_____	_____	_____
0	_____	_____	_____	9	_____	_____	_____

Notes: _____



20. Lung Auto PEEP

PEEP Test				Lung Auto PEEP				Pass	Fail
Lung compliance should be set to level 4. f: 13; V: 500 ml				Lung compliance should be set to level 4. PEEP: 0 mmHg; f: 13; V: 500 ml					
PEEP	Trial 1	Trial 2	Trial 3	Auto PEEP	Trial 1	Trial 2	Trial 3		
0	_____	_____	_____	0	_____	_____	_____		
5	_____	_____	_____	1	_____	_____	_____		
				2	_____	_____	_____		
				3	_____	_____	_____		

Notes: _____

21. Test Conclusion

Drain All Fluids 1. Fluid Reservoir (urinary) 2. Blood Reservoir (abdomen) 3. Tears/Sweat Reservoir (head)	Pass ☐ ☐	Fail ☐ ☐	Warning - Turn simulator OFF when removing or attaching limbs to the body. Remove and Re-attach Limbs	Pass ☐ ☐	Fail ☐ ☐
Simulator Shutdown 1. UNI app closes 2. Simulator shuts down	Pass ☐ ☐	Fail ☐ ☐		1. Left Arm 2. Right Arm 3. Left Leg 4. Right Leg	☐ ☐ ☐ ☐

Notes: _____

22. Accessory Box

Visual Inspection of Accessory Items	Pass	Fail
1. Items in Accessory Box (extra inserts and accessories)	☐	☐

Notes: _____

23. Observations

24. Inspection Information

Inspected by: _____ Date: _____

Document Information

Document No.: FORM-102238-EN-02	Edited by: Michel Alonso	Date: 12/13/22	Change(s): Rearranged test sequence and updated test requirements.
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